G – 12461

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BTech Degree Examination November 2024

Seventh Semester

Information Technology

20ITT71 – BLOCK CHAIN TECHNOLOGY

(Regulations 2020)

Time: Three hours Maximum: 100 marks

Answer all Questions

Part – A (10 × 2 = 20 marks)

1. Draw the blockchain generic structure. [CO1,K2]
2. State the CAP theorem in blockchain. [CO1,K1]
3. Distinguish between centralized system and decentralized system. [CO2,K2]
4. Specify the use of Eliptic Curve Cryptography (ECC) in blockchain. [CO2,K1]
5. Differentiate between fullnode and SPV node in bitcoin. [CO3,K2]
6. Show the equation for calculating difficulty target in creating a block in bitcoin. [CO3,K1]
7. List the important parameters are need to create an account in Ethereum. [CO4,K1]
8. Sketch the Hyperledger fabric application model. [CO4,K2]
9. Outline the unique properties of Hyperledger sawtooth lake. [CO5,K2]
10. Mention the ways to prevent financial crime by using blockchain. [CO5,K1]

Part – B (5 × 16 = 80 marks)

11. a. i) Name the different types of blockchain technology and elaborate the applications of it. (8) [CO1,K1]

    ii) Assume a scenario, where you do an online purchase of electronic gadgets from any online website using credit card. Make use of blockchain technology and show the steps while processing the above financial transactions. (8) [CO1,K1]

(OR)

    b. i) Outline the essential characteristics and their of blockchain technology with an example. (8) [CO1,K1]

    ii) Examine the challenges in block chain technology while adopting the E–Voting. (8) [CO1,K1]

12. a. i) Identify the elements in full ecosystem decentralization and describe in detail. (10) [CO2,K3]

ii) A company Y, a multinational technology firm in software development, adopted a decentralized organizational structure to enhance innovation and agility in response to market demands. Examine the entities in decentralized organization. (6) [CO2,K3]

(OR)

b. i) Define hashing. Discuss the use of hashing in block chain technology. (8) [CO2,K3]

ii) Suppose Alice wants to send the message "ERODE". Computers represent text as long numbers (01 for "A", 02 for "B" and so on). The RSA encryption scheme is used to encrypt the message. Alice uses two prime numbers 7 and 11 and public key is 31. (8) [CO2,K3]

13. a. i) Describe the different types of transaction carried out in the bitcoin. (8) [CO3,K2]

ii) Discuss the impact of adoption of crypto currencies in the banking and financial institutions. (8) [CO3,K2]

(OR)

b. Write the significance of prime coin's unique contribution to blockchain technology and how it might be useful in future blockchain innovations. (16) [CO3,K2]

14. a. Elaborate the operation of GasFee in Ethereum blockchain with neat diagram. (16) [CO4,K2]

(OR)

b. Describe the block header in Ethereum and different types of wallets used in Ethereum. (16) [CO4,K2]

15. a. With neat sketch, describe the principle and working of Hyperledger fabric also analyze the factors and choices involved when a bank considers implementing hyperledger Fabric. (16) [CO5,K3]

(OR)

b. You are working as chief Technology officer in passport office. You want to apply blockchain technology to maintain the documents of all the applications in a safe manner. How will you do that? Investigate in detail. (16) [CO5,K3]

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	24	41	35	–	–	–

Kongu Engineering College

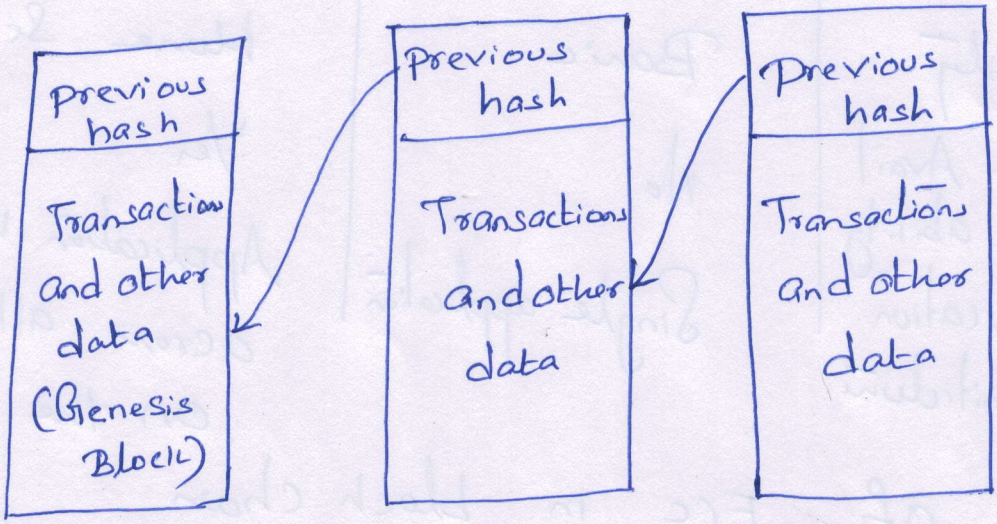
Department of IT

20ITT71 – Blockchain Technology

Part-A

(10 x 2 = 20 M)

1. Blockchain Generic Structure



2. CAP theorem

Consistency – that ensures that all nodes in a distributed System have a Single latest Copy of data.

Availability – System is up, accessible for use and is adepting incoming request

Partition tolerance – if a group of nodes fail the distributed System still Continues to operate Correctly.

2. Centralized System and decentralized System

Feature	Centralized	Decentralized
Ownership	Service Provider	All users
Architecture	client/server	Distributed, different topologies
Security	Basic	More Secure
High Availability	No	Yes
Application Architecture	Single application	Application replicated across all nodes on the network

4. use of Ecc in bloch chain

* Validate Menage
* Provide data integrity Services
* Smaller key size Provide greater Security.

5. Full node and Spv

Full node → bitcoin Core client Performing the wallet, miner, full blockchain storage and network routing fenctions

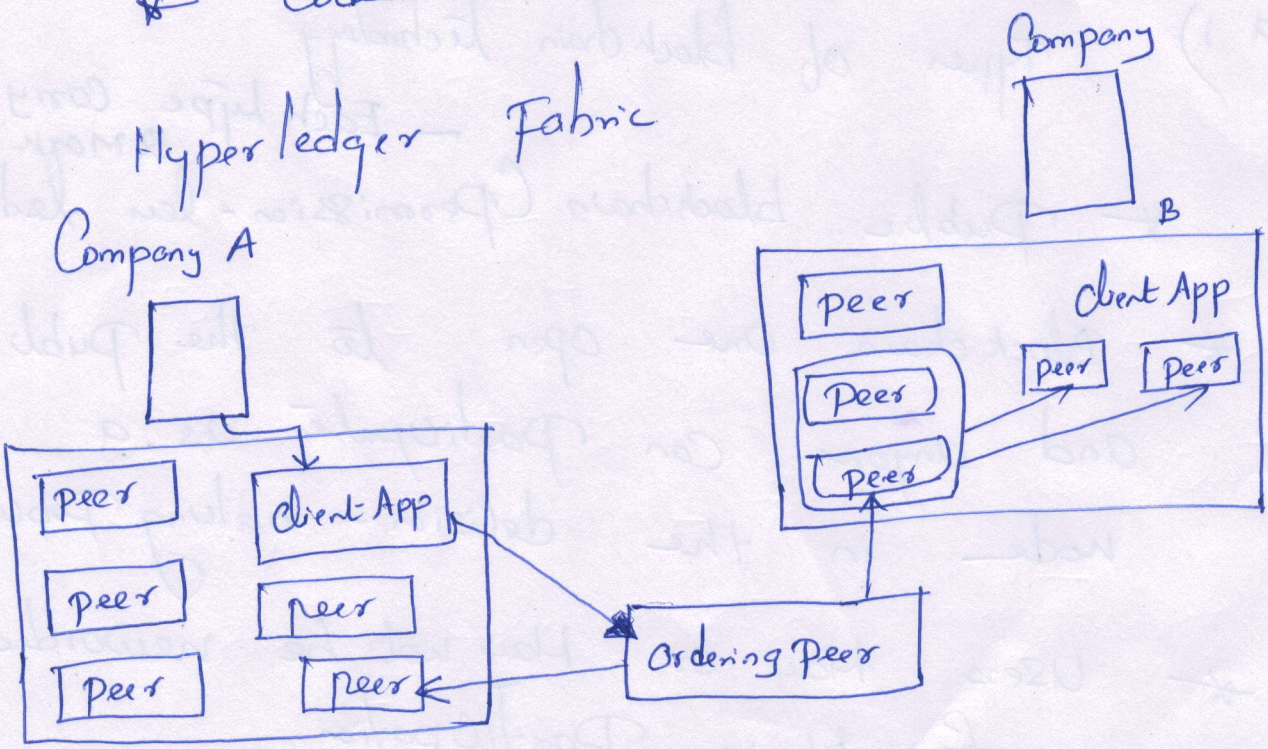
Spv nodes – light weight clients, perform only wallet and network routing functionality.

6. Difficult target in creating a block in block chain

$$Target = Previous\ target * time/2016 * 10\ minutes$$

7. Account state consist of four fields
   - * Nonce
   - * balance
   - * Storage root
   - * Code hash

8. Hyperledger Fabric



9. Properties of Hyperledger Sawtooth lake

Sawtooth lake can run in both Permissioned and non-Permissioned nodes

It introduces Proof of Elapsed Time (PoET) and the record idea of transaction families

10. Know Your Customer (KYC) and Anti Money laundering (AML) are the key enablers for the Prevention of financial Crime

Part-B

(8)

11) a i) Types of blockchain technology — Each type carry & More

* Public blockchain (permission-less ledgers)

* Blockchain are open to the public and anyone can participate as a node in the decision-making process.

* Users may or may not be rewarded for their participation

* All user of the Permissionless ledger Maintain a Copy of the ledger on their local nodes

Semi - Private blockchain

* Part of the blockchain is Private and Part of it is public.
* The Private Part is Controlled by a group of individuals whereas the Public Part is open for participation by anyone

Permissioned ledger

* A Permissioned ledger is a blockchain whereby the participants of the network are known and already trusted.
* do not need to use a distributed Consensus Mechanism.

Application of blockchain (2 M)

* Healthcare
* CryptoCurrency
* EVoting
* Supply Chain Management
* Personal identity Security.

(8)

11 a) 00 11) Steps for proceeing the above financial Transaction. Each step carry 2 M

* Transaction Details Preparation:

The website generates a detailed invoice with the product details and information.

* Smart Contract Activation:

A Smart Contract is created on the blockchain to Manage the transaction.

The Smart Contract includes

— Buyer & Seller wallet address

— Payment terms

* Blockchain Network Validates the transaction

Nodes in the blockchain network

* Validate the transaction by checking
  - the authenticity of the Customer's digital Signature
  - Sufficient funds in the Customer Wallet.

The Smart Contract releases the Payment from third Party to the seller's wallet upon Sucessful delivery Confirmation.

* The transaction is marked as Complete on the blockchain.

Both the Customer and Seller can access a transparent, immutable record of the transaction.

(8)

11 b i) Essential characteristics of blockchain technology

Characteristics (4M)

* DeCentralized
* Immutability
* Smart Contract

* Transparent
* Secure

For example Supply Chain Management (4M)

- A Supplier records the Production of raw Materials on the blockchain.
- Data such as origin, quality and Certifications are added to the blockchain.
- Manufacturer add records of Production Process
- Shipping Company stores transportation details such as departure
- Smart Contracts ensure Payments
- Retailers access blockchain records to verify Product
- Customers Provide feedback which is record on blockchain.

11b ii) challanges in blockchain while adopting (8M)

E- voting Each challenges carry (2M)

* Scalability
  - Scaling issues in public blockchain networks is primarily due to the restriction imposed by their own Consensus Mechanism.
* Cyber Security Threats
* Regulation and law
* Technological Complexity
* Privacy

12) a i) Full EcoSystem decentralization (10M)

* It is necessary that the environment around the blockchain is also decentralized.
* It include 3 elements (4M)

Storage -

Data can be stored directly in a blockchain
- not suitable for storing large amounts of data
- Distributed hash tables (DHTs)

(3M)

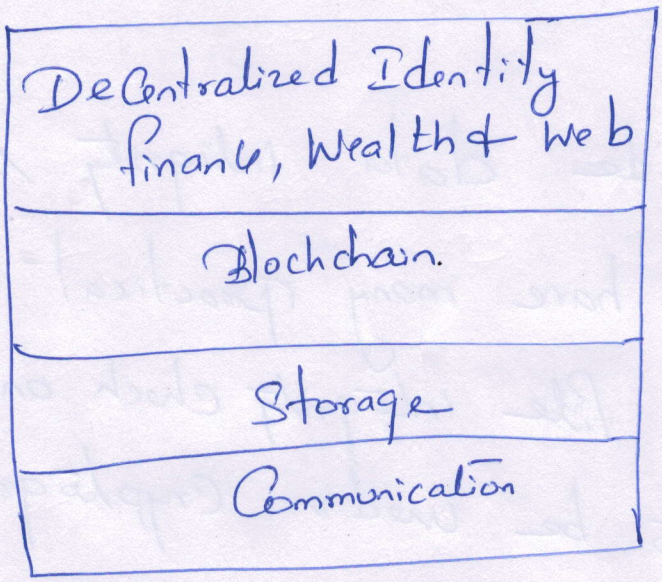
Communication

- The Communication layer in blockchain is decentralized
- Service such as e-mail and online storage in which service provider in control and users trust them.

(3M)

Computation (Computing power and decentralization)

- De Centralization of Computing or processing is achieved by a blockchain technology such as ethereum, where smart contracts with embedded business logic can run on the network.



DeCentralized Ecosystem overview

12a

(6M)

ii) Entities in decentralized organization

* DeCentralized organization (DOs) are software programs that run on a blockchain and are based on the idea of real human organizations with people & protocols. (2M)

* DeCentralized Autonomous organization (DAO)
  - fully automated (2M)

* DeCentralized Autonomous Corporation (DACs) (2M)
  - Make Money via shares offered to the participants and by paying dividends

12 b

(8M)

i) Hashing

(4M)

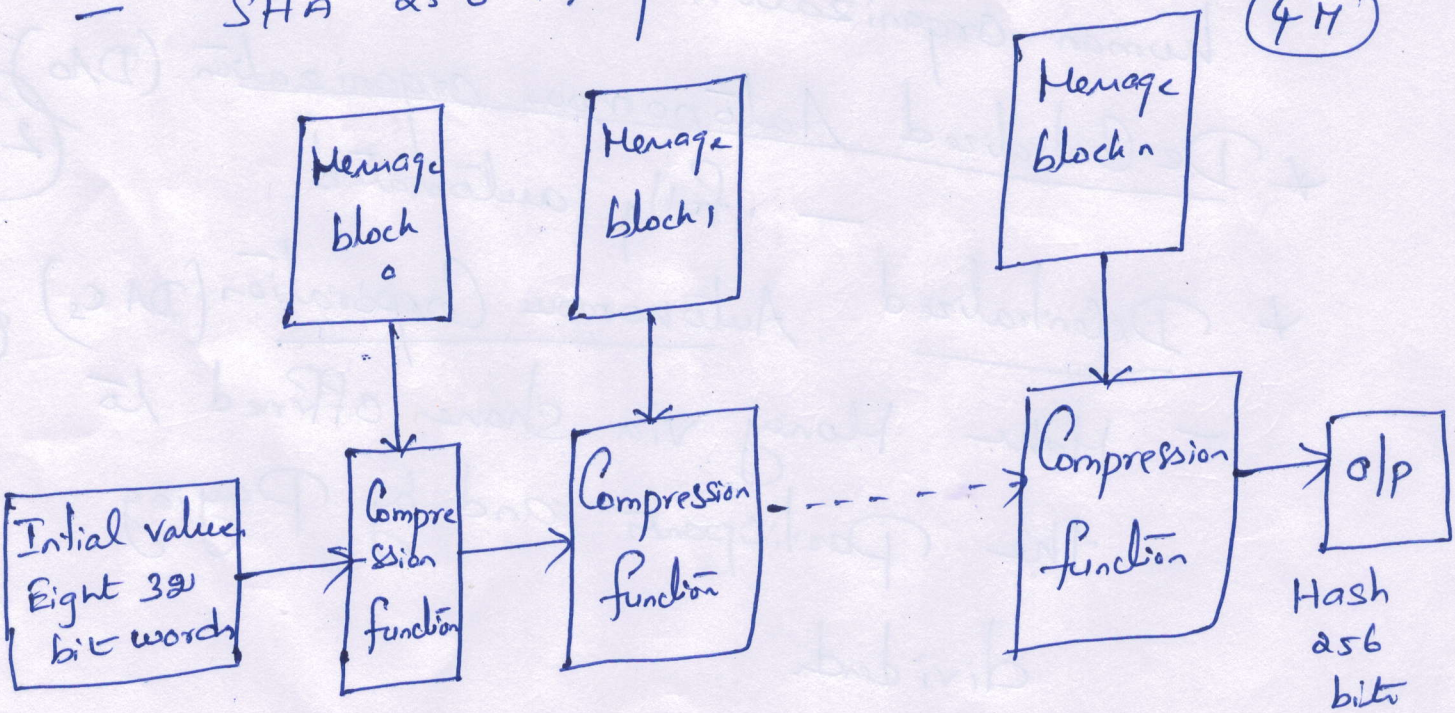
— Provide data integrity service

Hash function have many practical applications from simple file integrity check and password storage to be used in cryptographic protocols.

— They are used in hash table, distributed hash tables, bloomfilter, virus finger printing, Peer-to-peer P2P file sharing + Many other applications.

— SHA 256 Algorithm used in blockchain

(4M)



12 b

(8M)

ii) Erode

E ⇒ 5

R ⇒ 18

O ⇒ 15

D ⇒ 4

E ⇒ 5

Total = 47

So M = 47,

Given

$p = 7, \quad q = 11, \quad e = 31$

$n = p \times q$

$= 7 \times 11$

$= 77$

Encryption

$C = M^e \bmod n$

$= 47^{31} \bmod 77$

$= 7$

13 a. i) Different types of Transaction carried out in bitcoin (8M)

Various Scripts available in bitcoin to handle the value transfer from the source to destination.

* Pay to Public key hash (P2PKH): It is used to send transactions to the bitcoin address. (3M)
The Script pubkey and Scriptsig Parameters are Concatenated together and executed.

* Pay to Script hash (PSH): is used in order to send transaction to a Script hash (that is the address starting with 3) and was standardized in BIP16. (2M)

* Multi Sig (Pay to Multi sig): M of n multi Signature transaction is a Complex type of Script where it is possible to Construct a Script that required multiple signatures to be valid in order to redeem a transaction. (3M)

ii) Impact of adoption of Crypto Currencies in banking and financial institution (8M)

* Faster transaction
* low Transaction fee
* Global Accessibility
* Decentralized financial services
* Transparent
* Allowing institutions to enage in DeFi

(16M)

13 b) Prime Coin - Contribution to blockchain technology

- first digital currency on the Market that introduced a useful PoW, Primecoin uses Searching Prime numbers as a PoW.
- Three types of prime numbers (Cunningham, Cunningham, bitwin)
  - ↓ first kind
  - ↓ second kind

- The difficulty is dynamically adjusted via a continuous difficulty evaluation scheme
- efficient Verification of PoW based on Prime numbers in also of high importance
- Prime chain are selected as a PoW because finding Prime chains gets difficult as the chain increases in length whereas Verification remain quick enough to warrant being used as an efficient PoW algorithm.
- PoW Certificate is produced by linking the Prime chain to the block header hash
- It also requires that the block header Origin is divisible by the block header hash.
- Adjustable difficulty of PoW algorithm is met by introducing difficulty

adjustment every block instead of every 2016

— Smoother approach as Compared to bitcoin and allows readjustment in the case of sudden increases in hash power.

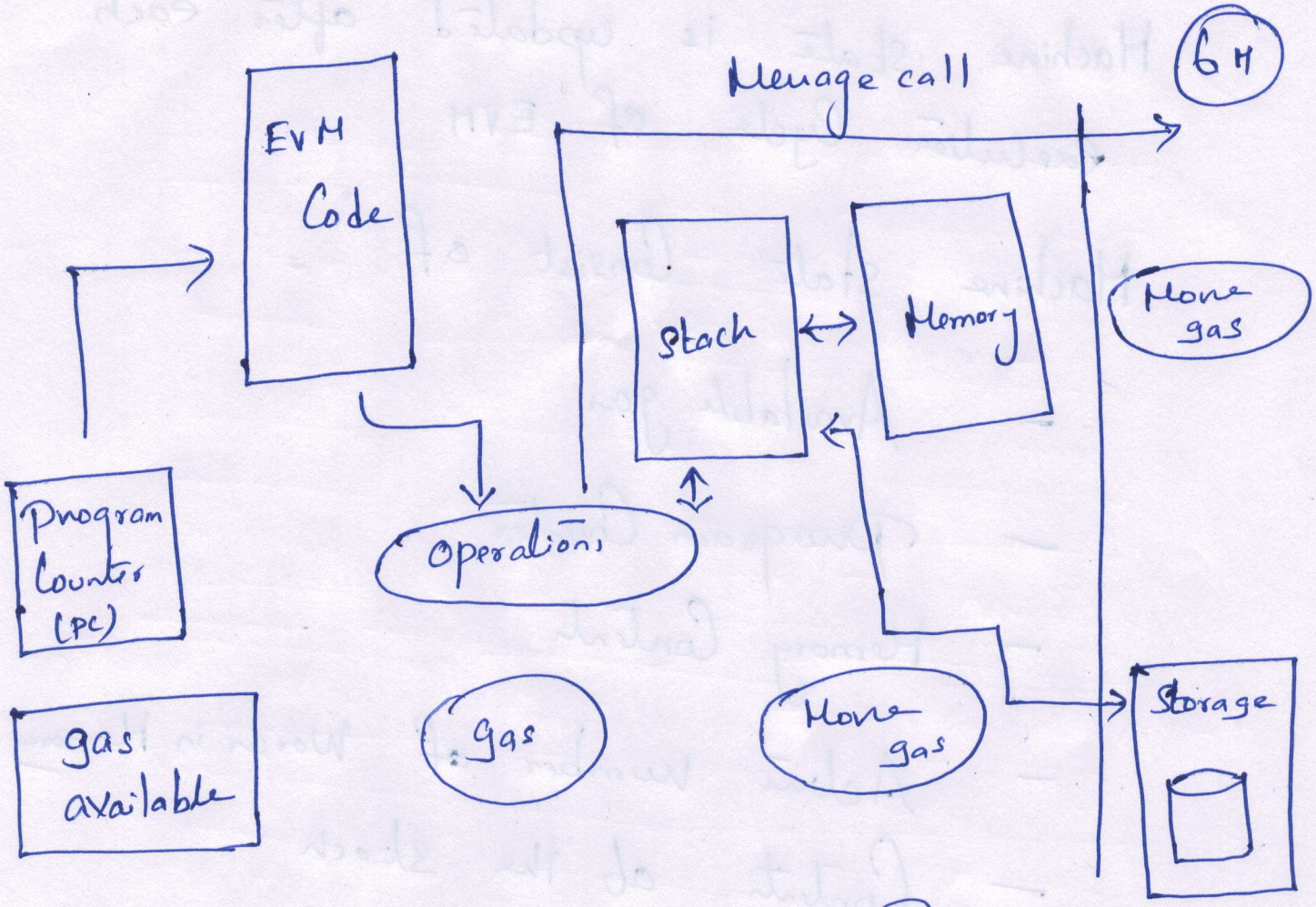
— there is no Concrete limit on the number of Coins Primecoin Can generate

14) a Operation of Gas fee in Ethereum (16M)

— EVM (Ethereum is a Simple stack based (18M) execution machine that runs bytecode instructions inorder to transform the System state from one state to another.

— EVM is a turing Complete machine but is limited by the amount of gas that is required to run any instruction

— Program Counter and EVM stack are updated accordingly with each instruction execution



In EVM where the Virtual ROM stores the Program code that is Copied into main Memory using CodeCopy.

The main Memory is then read by the EVM by referring to the Program Counter and executes instruction step by step.

Machine state is also Maintained internally by the EVM.

Machine state is updated after each execution cycle of EVM

Machine state consist of

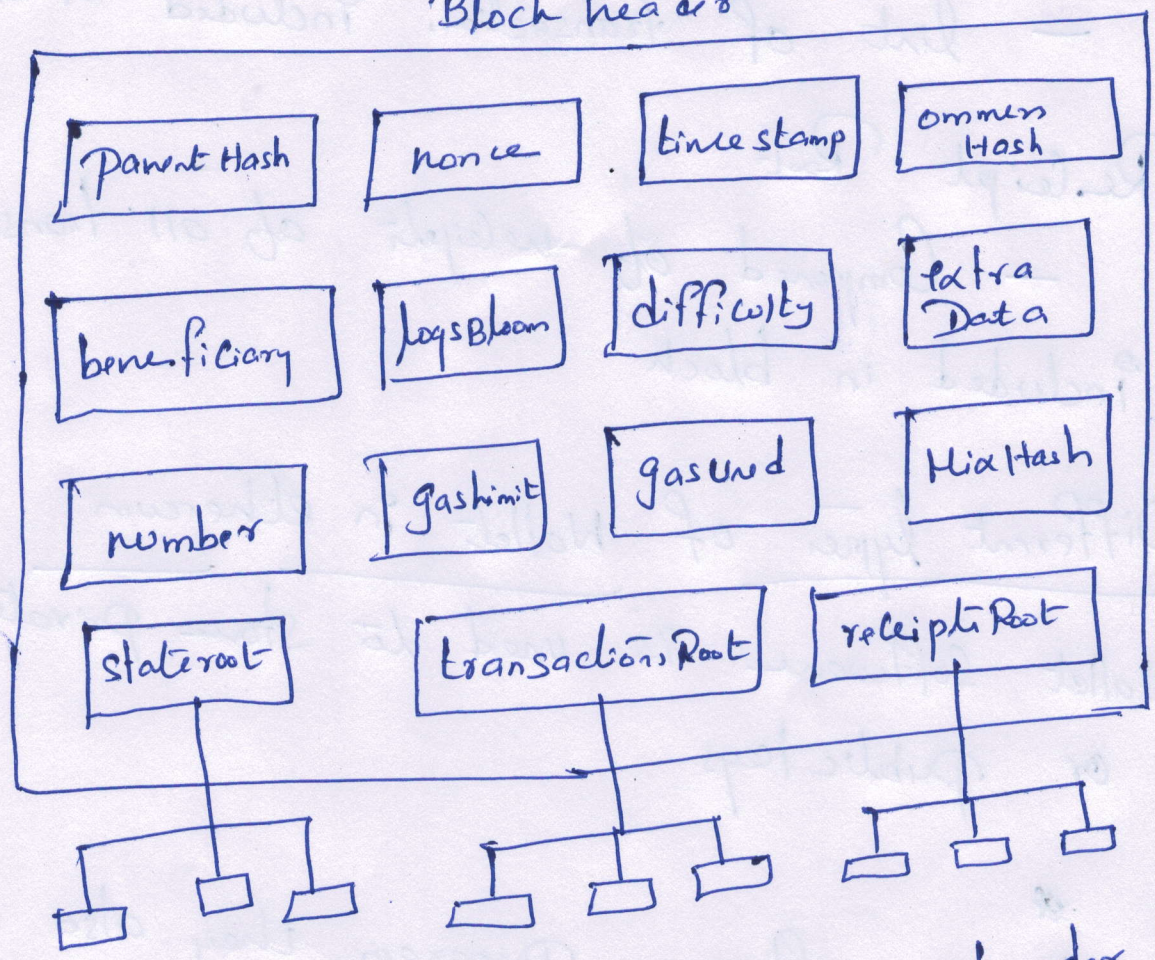
- Available gas
- Program Counter
- Memory Contents
- Active number of Words in Memory
- Contents of the stack

Ethereum gas fees are the transaction fees user pay on the ethereum blockchain to conduct transaction and execute Smart Contract.

If the execution is Sucessful, then the account is created after the payment of appropriate gas Cost.

Exact price of the gas is determined by Supply, demand and network capacity at the time of the transaction.

14 b) Block header in Ethereum (16M)



(4M)

(6M)

Parent hash – Previous block header

OMMers hash – list of uncle block included in the block

Beneficiary – that will receive the mining reward once the block is successfully Mined

State root
- → 256 bit hash value
- → It is calculated after all transaction have been processed and finalized.

Transaction Root
- list of transactions included in block.

Receipt Root
- Compared of receipts of all transactions included in block

Different types of Wallets in ethereum

- Wallet software is used to store private or public keys

- As a software program, they also provide some functions to the users to manage and carry out transactions on the bitcoin network (6M)

- Non-deterministic Wallet
- Deterministic Wallet
- Hierarchical deterministic
- Brain Wallet
- Paper
- Hardware
- Online

Mobile Wallet

↳ are available for the Android Platform and ios, for example bread Wallet, Copay & Jaxx.

16 M

15 a) Hyperledger Fabric

— Enable a modular, open and flexible approach towards building blockchain networks

12 M

— Various functions in the fabric are pluggable, and it also allows use of any language to develop smart Contract.

— Chaincode (smart Contract) is Sandboxed into a Secure Container which includes a Secure operating System, chaincode language

— It is a Permissioned network which address issues such as scalability, Privacy and Confidentiality

— all Participants are required to be registered with the Membership Service

inorder to acces the blockchain network

- It include membership Services, blockchain Services and chaincode Services

Membership Services

— user identity validation and user registration

Blockchain Services

— Consensus manager is responsible for providing interface to the Consensus algorithm

Chaincode Services

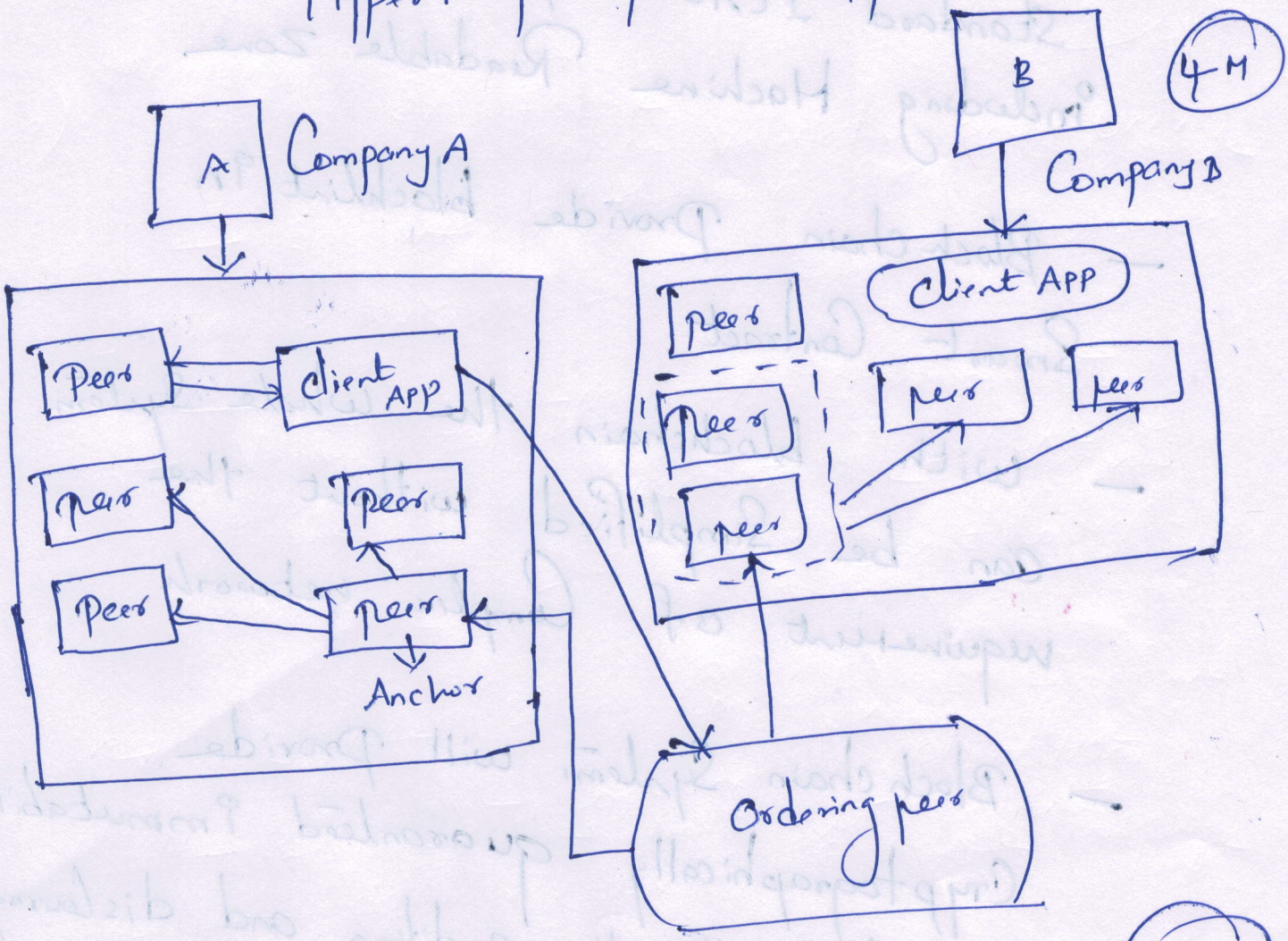
— Service allow the creation of Secure Containers that are used to execute the chaincode.

Secure Container

— chaincode is deployed in Docker Container that provide a locked down Sandboxed environment for Smart Contract execution.

Selone registry
— provides a record of all images containing Smart Contracts

Hyperledger Fabric Application Model



16M

15b) Apply blockchain technology to maintain documents of all the applications in a safe manner — 12M

— Machine readable travel documents and specifically biometric passports

have paved the way for automated border Control

- Biometric features such as retina scan, finger prints, facial recognition and Standard ICAO specified features including Machine Readable Zone

- Blockchain provide blocklist in Smart Contract

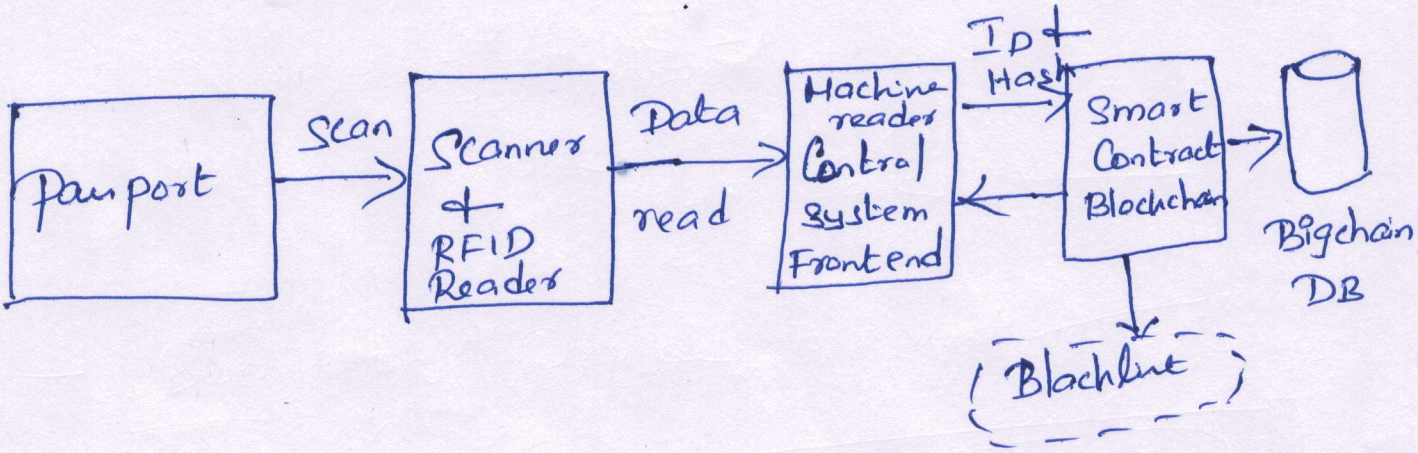
- with blockchain the whole system can be simplified without the requirement of Complex network

- Blockchain systems will provide Cryptographically guaranteed immutability which helps with auditing and discourages any fraudulent activity

- Passport scanning to an RFID and Page Scanner, which reach the data page and extract machine readable

information along with a hash of the biometric data stored in RFID chip.

(4 M)



These participants can be worldwide Consortium of homeland security departments of various nations.